

A Deep Reinforcement Learning-Based Feature Selection Method for Invasive Disease Event Prediction Using Imbalanced Follow-Up Data

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Abstract—The machine learning-based model is a promising paradigm for predicting invasive disease events (iDEs) in breast cancer. Feature selection (FS) is an essential preprocessing technique employed to identify the pertinent features for the prediction model. However, conventional FS methods often fail with imbalanced clinical data due to the bias towards the majority class. In this paper, a novel FS framework based on reinforcement learning (RLFS) is developed to identify the optimal feature subset for the imbalanced data. The RLFS employs an iterative methodology, wherein data resampling technique generates a balanced dataset before each iteration. A decision network is trained using a deep RL algorithm to identify the relevant features for the dataset in the current iteration. With such an iterative training strategy, numerous constructed datasets gradually boost the FS capacity of the decision network, resulting in a robust performance for imbalanced data. Finally, a weighted model is proposed to determine the most suitable FS solution. The RLFS is employed to predict breast cancer iDEs using real follow-up data. The comparison results demonstrated that RLFS effectively reduces the number of features while outperforming several state-of-the-art FS algorithms.

Index Terms—Deep reinforcement learning, feature selection, breast cancer, invasive disease events, imbalanced data.

I. INTRODUCTION

THE high morbidity and mortality rates associated with breast cancer pose a significant threat to the overall health

of women worldwide [1]. Although the combination of breast cancer surgery and adjuvant therapies represents a practical approach to preventing the spread of cancer [2], patients remain susceptible to the occurrence of invasive disease events (iDEs) at any time, including recurrence, metastasis, contralateral tumors, and secondary malignancies [3]. Therefore, there is a significant demand for early prediction of iDEs following the primary tumor, which could facilitate timely intervention and enhance patients' quality of life. In recent years, the rapid advancement of artificial intelligence technology in the biomedical field has sparked a growing interest in developing clinical decision support systems by machine learning (ML) methods with extensive patient clinical records [4], [5], [6].

However, the ML-based medical diagnosis models encounter two significant challenges arising from clinical data: high dimensionality and class imbalance. The term “high dimensionality” refers to the presence of certain features that may not necessarily contribute to the performance of the ML model [7]. The data may contain numerous correlated, redundant, or even irrelevant features. These features typically impede the speed of learning a model and can potentially diminish its accuracy [8]. The class imbalance leads to the model focusing more on the majority class [9]. Consequently, the model is biased towards the majority class, resulting in incorrect diagnoses for the minority class. However, misclassifying an instance from the minority class incurs higher costs than misclassifying instances from the majority class. In this context, numerous feature selection (FS) algorithms and data balancing techniques have been developed to tackle these two challenges independently.

In order to address the high dimensionality problem, a variety of FS methods have been developed, which can be categorized into three groups based on different search strategies: global optimal search methods, evolutionary computational (EC) methods, and heuristic search methods [10]. Global search methods employ enumeration or branch and bound algorithms to search for the global optimal solution [11]. The EC-based methods can efficiently generate near-ideal solutions in an acceptable time using evolutionary algorithms, such as genetic algorithm (GA) [12] and particle swarm optimization (PSO) [13]. The

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heuristic search methods rely on the heuristic rule to construct a feature subset, such as the sequential forward-selection (SFS) [14] and sequential backward-selection (SBS) algorithms [15]. The use of heuristic search in practical applications is widespread due to its high efficiency. While applying the above FS methods can significantly improve training efficiency, accuracy, and interpretability of downstream ML models [16], it is important to note that these methods have primarily been developed for general FS problems and may not be suitable for scenarios involving imbalanced data. The evaluation of feature subsets for FS methods is predominantly influenced by the majority classes, leading to a biased selection of features [17].

On the other hand, the research on data imbalance was conducted independently of the FS methods [9]. Specifically, the resampling technique is widely acknowledged as an effective approach in addressing class imbalanced problems. This technique operates by adding or removing the samples within the data space to mitigate the biases associated with imbalanced datasets, such as over-sampling [18] and under-sampling [19]. While considerable research focuses on the issue of class imbalance, there remains a lack of exploration of integrating data balancing techniques within FS algorithms. Most studies predominantly employ a straightforward approach, which involves initially employing data balancing techniques to construct a balanced dataset, followed by FS on this dataset [20]. However, the inherent randomness in data balancing techniques leads to variability in the outcomes of each execution, significantly compromising the robustness and stability of the algorithm. Despite the inherent similarity of the datasets being constructed by data balancing techniques, traditional algorithms fail to leverage this characteristic systematically.

To overcome the aforementioned problems, this paper proposes an innovative FS framework that can effectively address the challenges posed by high-dimensional and imbalanced data in breast cancer iDEs prediction problem. This framework contains a decision network based on deep neural networks. The decision network is trained using the reinforcement learning (RL) algorithm in a self-supervised manner [21] to select the most valuable features for the given dataset. The decision is made based on the performance of the selected features and statistics of the given dataset. The main distinction between this framework and previous FS research lies in adopting an iterative incremental learning methodology. Before each iteration, a balanced dataset is constructed using resampling technique, and then the decision network is trained on the constructed dataset. Each constructed dataset can be viewed as a hypothetical space in the real data space. Therefore, the decision network can gradually acquire robust FS capabilities that approximate the true space during the iterative learning process. Specifically, the contributions of this paper are outlined below:

- 1) The FS problem is formulated within the framework of reinforcement learning. A deep neural network acts as an agent for decision-making, selecting actions based on the observed state, where the state represents the given dataset, and the action determines which feature to select. The agent is guided by a designed reward signal to generate a feature set with superior performance.

- 2) An iterative training method is proposed, where a balanced dataset is constructed by resampling technique before each iteration for training the agent. Specifically, a graph convolutional network-based (GCN) state representation method is proposed to transform the dynamically changing dataset into a fixed-dimensional state, which serves as the input for the decision network.
- 3) After completing the iteration, a weighted model is proposed to identify final features that exhibit relevance across a significant proportion of iterations, thereby mitigating the influence of randomness from the resampling technique.
- 4) Extensive experiments are conducted based on a clinical cohort with 2645 breast cancer patients from Xiangya Hospital of Central South University, China, to demonstrate the enhanced performance of the proposed method.

II. RELATED WORKS

A. Machine Learning in Breast Cancer Prognosis

The development of clinical information collection technologies facilitates the applications of ML in breast cancer prognosis. This section provides a brief overview of several recent works in this field.

Massafra et al. [22] proposed an ensemble learning method to tackle the task of breast cancer iDEs prediction. The proposed method combined several foundational models to generate a final prediction through the voting mechanism. Each model on the training data employed an embedding FS method via the Boruta technique. The experimental results demonstrated that using ensemble learning techniques can significantly enhance the performance of the initial basic models. However, due to the necessity of repeatedly training multiple FS algorithms, the computational complexity associated with ensemble learning surpasses that of individual algorithms.

Fu et al. [23] proposed a prognostic model for predicting invasive disease-free survival (iDFS) in early-stage breast cancer patients. The proposed model employed the stratified FS method to identify the significant features from the original data. In the first stage, a statistical FS (SFS) method was employed to generate a reduced dataset of elite features. Then, the ensemble feature selection based on the XGBoost algorithm was employed to determine the final feature subset. The method indicates that FS can reduce ML model complexity and enhances model interpretability and comprehensibility. However, determining an optimal threshold for the SFS method remains a persistent challenge.

Teng et al. [24] developed a Bayesian Dynamic Cox (BDCox) model to accurately predict the 5-year overall survival (OS) of breast cancer using highly imbalanced follow-up clinical data. The most relevant features were identified using three FS methods: fast backward variable selection (fastbw), Elastic net, and Bayesian model average (BMA). The experimental results demonstrated that the features selected by BMA and fastbw exhibit superior accuracy compared to the baseline model, while Elastic net fails to outperform the baseline model. It can be seen

that the presence of imbalanced data introduces bias into certain FS algorithms and even deteriorates model performance.

Juwara et al. [25] conducted a systematic investigation of neuropathic pain in patients following breast cancer surgery using ML. They employed six FS methods to identify significant pain features that can aid in determining the susceptibility of patients to neuropathic pain. The findings of this study suggest that ML models exhibit superior predictive performance compared to traditional least square contexts of neuropathic pain. However, the authors of this paper also noted that the imbalanced dataset resulted in insufficiently robust training features.

The works mentioned above highlight the significance of FS methods and ascertain that they can substantially enhance the prediction performance of breast cancer prognostic models. However, the majority of works appear to ignore the issue of data imbalance, which is a well-established issue in realistic breast cancer diagnoses due to the limited number of recurrent patients. Therefore, developing a general and systematic FS method for imbalanced data represents an urgent problem that necessitates resolution.

B. Feature Selection for Imbalanced Data

Recently, researchers have been exploring cost-sensitive approaches and re-sampling techniques to address challenges associated with FS for imbalanced datasets [9].

Cost-sensitive approaches involve modifying the learning objective to construct a classifier that can effectively adapt to imbalanced data. Moayedikia et al. [26] proposed a novel feature selection method based on symmetrical uncertainty (SU) for imbalanced data. The SU measure was employed to assess the features of class labels, aiding in identifying influential features for retrieving infrequent class labels. Zhang et al. [27] proposed an enhanced F-measure method based on filling risk (RF-measure) to assess the impact of missing data on the performance of FS in imbalanced datasets. Then, a PSO-based FS method with fuzzy clustering is proposed based on RF-measure. These cost-sensitive approaches are computationally effective techniques. Nevertheless, such approaches compel FS algorithms to prioritize the minority class. This is often ineffective in practice, as most samples in the dataset predominantly originate from the majority class. This imbalance can result in erroneous compensatory outputs.

The basic idea behind the re-sampling technique involves iteratively adding, removing, or synthesizing samples until balanced data are achieved. Yin et al. [17] proposed a class decomposition-based FS approach, which involves partitioning major classes into smaller pseudo-subclasses and generating corresponding pseudo-class labels. The joint impact of feature selection and data re-sampling on imbalanced data was systematically investigated by Zhang et al. [20]. The findings indicate that adopting the hybrid framework should be considered when seeking the optimal imbalanced classification model. Random sampling is simple but insufficient in many cases due to the randomness, which can result in an uneven distribution of samples across different parts of the dataset. Ensemble learning can alleviate

such problems by continually generating samples and combining them to enhance performance. For example, the Bagging method involves partitioning original data into multiple sets of equal size [9]. Each, consisting of the same number of instances, is used to train an individual classifier. The aggregation of these classifiers results in a composite classifier.

In summary, the prevailing approaches to FS for imbalanced data are generally simplistic, typically involving initial data balancing to create a balanced dataset, followed by FS on this modified dataset. Although these methods partially mitigate the feature selection bias caused by data imbalance, there are still outstanding issues that require resolution. On the one hand, the true data space cannot be fully captured by any hypotheses space generated through a single sampling process. FS methods may produce feature subsets that can be regarded as local optima within the true space. On the other hand, the inherent randomness associated with data balancing techniques results in varying outcomes across different runs, significantly undermining the algorithm's stability.

III. PROBLEM FORMULATION

A. Feature Selection Problem

Consider a dataset D containing m samples and n features $X = \{x_i, i = 1, 2, \dots, n\}$. The objective of the FS is to identify a feature subset $S = \{x_1, x_2, \dots, x_{|S|}\}$, $S \subseteq X$, which can maximize the classification accuracy while minimizing the number of features required.

Furthermore, the state of the feature subset can be generally represented as a binary encoding vector:

$$\mathbf{v} = (v_1, v_2, \dots, v_n), v_i \in \{0, 1\}, i = 1, 2, \dots, n, \quad (1)$$

where $v_i = 1$ denotes the selection of the i th feature x_i , while $v_i = 0$ indicates its exclusion. Therefore, the FS can be cast as a typical combinatorial optimization (CO) problem and formulated as following:

$$\begin{aligned} \mathbf{v}^* &= \operatorname{argmax} Acc(\mathbf{v}, D) \\ \text{s.t. } &\begin{cases} \mathbf{v} = (v_1, v_2, \dots, v_n), v_i \in \{0, 1\}, i = 1, 2, \dots, n, \\ 1 \leq \|\mathbf{v}\|_0 \leq n. \end{cases} \end{aligned} \quad (2)$$

where $Acc(\cdot)$ denotes the classification accuracy corresponding to the feature subset $\mathbf{v}$, and $\|\cdot\|_0$ indicates the number of the selected features in $\mathbf{v}$.

The above CO problem is NP-hard and lacks an efficient polynomial-time solution. In recent years, a prominent trend has emerged in addressing CO problems by applying RL algorithms [21]. In particular, RL demonstrates a notable capacity for generalization across a specific class of problems. For example, the RL algorithm can be trained on a set of already solved FS instances to determine the optimal features for a new instance. Consequently, FS is formulated as an RL problem in this paper. In the proposed iterative learning framework, the efficacy of FS is enhanced through the continuously generated datasets.

B. Reformulation as Markov Decision Process

Markov decision process (MDP) offers a widely adopted mathematical framework for modeling RL problems [28]. MDP can be defined as a tuple $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{T}, \gamma \rangle$, which represents the state space, action space, reward, transition function, and discount factor, respectively. Specifically, at any discrete time step t , the agent executes an action a_t according to policy π based on its observed state s_t . Then, the environment is transitioned to the next state s_{t+1} according to the transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \mapsto \mathcal{S}$. Finally, the agent receives an immediate reward r_{t+1} based on the reward function $\mathcal{R} : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$. The agent's target is to maximize its accumulated expected reward $G = \mathbb{E}_{s \sim \mathcal{T}, a \sim \pi} \left[\sum_{t=0}^T \gamma^t r_{t+1} \right]$, where $\gamma \in [0, 1)$ is the discount factor and T is the time step horizon. The components of the MDP for FS problem are described as follows.

- *Environment*: The environment is responsible for interacting with the agent and providing rewards as feedback. This paper uses the resampling technique to design a dynamic environment. The environment is reset at the beginning of each episode by generating a balanced dataset D^- from the original imbalanced dataset D using the under-sampling method. The agent chooses actions to select or deselect features based on the constructed dataset. Each episode's different datasets can be considered analogous problem instances emerging within the same domain. The generalization ability and robustness of the agent in feature selection will be enhanced through continuous learning.
- *State*: The agent's action is determined by its observation of the current state of the environment s_t . The state comprises three observations that the agent can observe:

$$s_t := \langle o_t^1, o_t^2, o_t^3 \rangle, \quad (3)$$

where o_t^1 denotes the sampled data D^- in this episode; o_t^2 represents the current decision time step; o_t^3 denotes the currently selected feature subset. This paper proposes a GCN-based method to transform the constructed dataset into a fixed-dimensional vector, which serves as the input for agent observation o_t^1 . The binary encoding vector $\mathbf{v}_t$ can naturally serve as the representation of the selected feature subset o_t^3 .

- *Action*: In this paper, the agent in the environment orderly determines the selection of each feature one by one. Therefore, at time step t , the action a_t involves either selecting ($a_t = 1$) or deselecting ($a_t = 0$) the current feature x_t .
- *Reward*: The reward function $r(s_t, a_t)$ in this paper is defined as the difference in the accuracy performance after executing an action a_t and transitioning to the next state s_{t+1} . That can be described as follows:

$$r(s_t, a_t) = \text{Acc}(\mathbf{v}_{t+1}, D^-) - \text{Acc}(\mathbf{v}_t, D^-), \quad (4)$$

where $\mathbf{v}_t$ denotes the feature subset at time step t ; $\mathbf{v}_{t+1}$ represents the changed feature subset after executing the action a_t . As a result, the cumulative reward precisely aligns with the value of the objective function in (2).

- *Policy*: The agent takes an action a_t according to its policy $\pi(a_t|s_t)$ based on the state s_t . In this paper, a neural

network Q is constructed to estimate the value of the action. The policy of the agent is determined by

$$\pi(a_t|s_t) := \operatorname{argmax}_{a_t \in \{0,1\}} Q(s_t, a_t). \quad (5)$$

IV. PROPOSED METHOD

This paper addresses the problem of FS for imbalanced data using a unique combination of RL algorithm and data resampling technique. The decision network Q constructs an FS solution based on the input dataset. An iterative training method is proposed to generate numerous balanced datasets. In the iterative process, the stability and robustness of the decision network in FS will be enhanced through training on diverse datasets. However, several challenges arise toward this goal. First, utilizing a dataset directly as input for a neural network poses significant challenges. What methods can be employed to transform it into an appropriate format? Second, in what ways can a neural network facilitate the FS process? Finally, how can the parameters of a decision network be effectively trained using the RL algorithm? This section provides a comprehensive introduction to the method proposed.

A. Framework Overview

The framework of the proposed reinforcement learning-based feature selection method (RLFS) is illustrated in Fig. 1. The framework mainly consists of two parts, i.e., the interaction part (green line) and the training part (yellow line). At the beginning of each episode (iteration), a balanced dataset is generated utilizing the random under-sampling technique. Then, the agent (Q-network) executes actions a_t to identify relevant features based on its observations s_t (i.e., the given dataset). To enable the dataset to function as input for decision-making within the decision network, a graph-based state representation method transforms the dataset into a fixed-length one-dimensional vector. ML models are used to evaluate the feature selection results and return classification accuracy as a reward signal to guide the agent's actions. Finally, the interactive data during a time slot $(s_t, a_t, r_{t+1}, s_{t+1})$ will be stored in the replay buffer to train the decision network. During the training phase, the agent leverages data stored in the replay buffer to refine the parameters of the Q-Network based on the DQN algorithm.

B. State Representation on Graph

The decision network Q identifies pertinent features derived from the provided dataset. However, a matrix representation of the dataset is unsuitable input for a neural network. Specifically, the environment dataset varies in each episode due to the resampling technique. The characteristics of the dynamic dataset are challenging to describe using a closed form and may rely on complicated statistics. This paper proposes a graph convolutional network-based (GCN) state representation method to capture intricate high-order information within the dataset. The proposed state representation method consists of two steps, as depicted in Fig. 2. The first stage involves acquiring compact representations for features, while

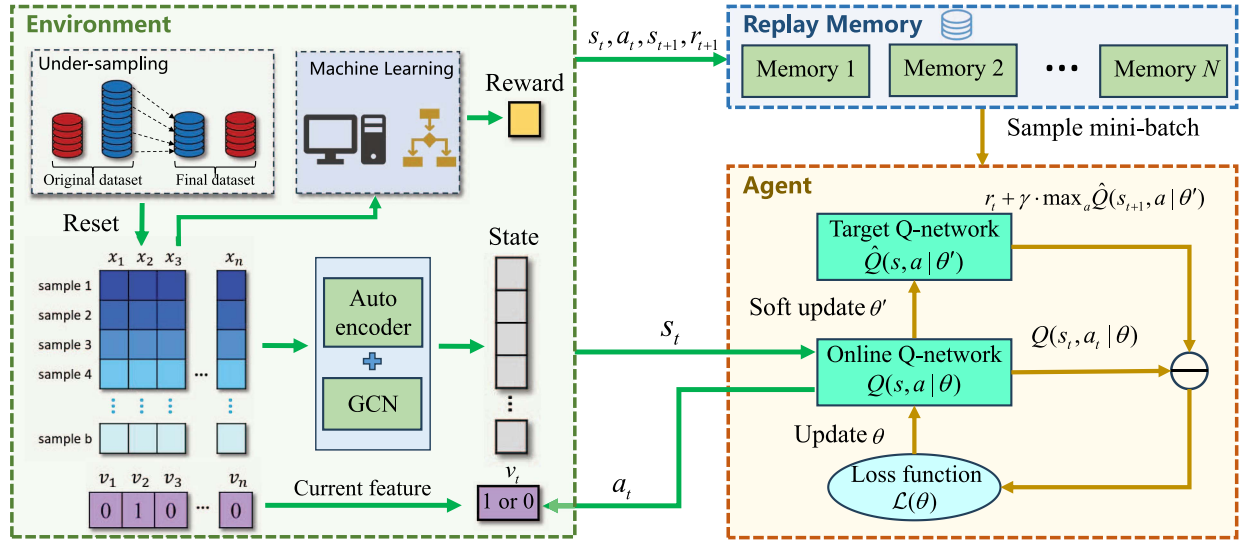


Fig. 1. The workflow of proposed method. In each epoch (iteration), a balanced dataset is first sampled from the original dataset. The auto-encoder and GCN techniques map the dataset into a fixed-length representation as the agent's observation. A decision network (agent) is designed to select features orderly based on the observation. The ML model is trained based on the selected features and returns a reward signal. The interaction data accumulated in the replay memory is used to train the parameters of the decision network by RL algorithm.

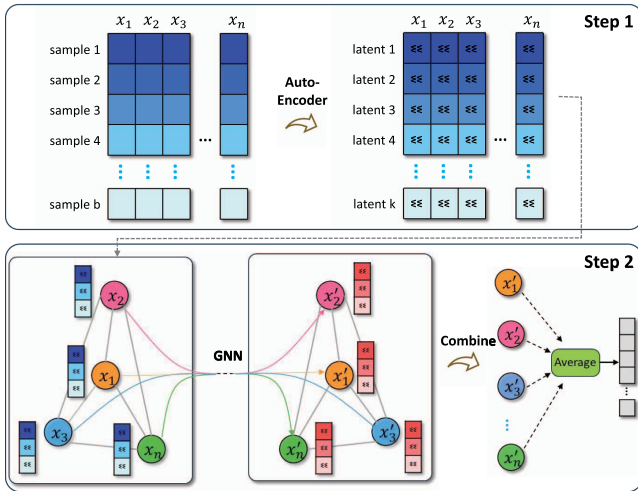


Fig. 2. The architecture of state representation method. In step 1, the auto-encoder reduces the dimension of the features from b to k , thereby mitigating the input dimension curse of the GCN. In step 2, the GCN captures the relationship among features and transforms the data matrix into a fixed-length state representation.

the second stage combines them to obtain a fixed-length state representation.

1) Autoencoder Based Data Dimensionality Reduction: The auto-encoder is widely utilized for representation learning, where it takes high-dimensional data as input and generates a low-dimensional latent representation by minimizing a reconstruction loss between the original and reconstructed data [29]. Given the current sampled dataset $D^- \in \mathbb{R}^{b \times n}$, which includes b data samples and n features, the auto-encoder transforms each feature column in D^- into a latent representation of length k ($k < b$), thereby obtaining a matrix $M \in \mathbb{R}^{k \times n}$. This step aims to generate a concise embedding for each feature, which serves

the purpose of mitigating the issue of high input dimensionality in the GCN.

2) GCN-Based State Representation: The auto-encoder technique enables the extraction of distinct latent properties for individual features while disregarding the intricate interrelationships among them. To this end, the matrix $M \in \mathbb{R}^{k \times n}$ is embedded into a complete graph G to represent their relationships, with each node representing a feature representation extracted by an auto-encoder. This method enables efficient generalization of the interconnections within the sampled dataset. In addition, the GCN, which represents the state-of-the-art graph embedding network [30], is employed to capture the characteristics of a node within its graph neighborhood. The layer-wise propagation rule of the multi-layer GCN can be expressed as follows:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right), \quad (6)$$

where $\tilde{A} = A + I_N$ is the adjacency matrix of the graph G with added self-connections, I_N is the identity matrix, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, $W^{(l)}$ represents a layer-specific trainable weight matrix, and $\sigma(\cdot)$ denotes an activation function. $H^{(l)}$ is the output matrix of the l^{th} layer, where $H^{(0)} = M$.

According to the update formula in (6), the embedding process is carried out based on the graph topology. Moreover, as more iterations are carried out, nodes propagate further and get non-linearly aggregated at distant nodes. Finally, after terminating at L iterations, the final latent representations $H^{(L)} \in \mathbb{R}^{k \times n}$ are obtained (middle of step 2 in Fig. 2). A k -length vector o_t^1 is generated by averaging the representation of each feature (right side of step 2 in Fig. 2).

C. Neural Network Based Feature Selection Process

In this paper, a neural network serves as the agent that selects actions a_t based on its observed state s_t . Specifically, feature

Algorithm 1: Neural Network Based Feature Selection Process.

Input: Dataset: D^- ; number of feature: n ; binary encoding $\mathbf{v} \in \mathbb{R}^n$.

- 1 Initialize binary encoding $\mathbf{v}$ with zero;
- 2 Initialize $o_1^1 \leftarrow$ representation of D^- generated by the GCN-based method, $o_1^2 \leftarrow 1$, $o_1^3 \leftarrow \mathbf{v}$;
- 3 $s_0 \leftarrow$ concatenate $\langle o_1^1, o_1^2, o_1^3 \rangle$;
- 4 Accuracy $\leftarrow 0$;
- 5 **for** $t = 1$ **to** n **do**
- 6 $v_t = \begin{cases} \text{random choice from } \{0, 1\}, & \text{rand} < \epsilon \\ \text{argmax}_{a_t} Q(s_t, a_t), & \text{otherwise} \end{cases}$;
- 7 $r_{t+1} \leftarrow \text{Acc}(\mathbf{v}, D^-) - \text{Accuracy}$;
- 8 Accuracy $\leftarrow \text{Acc}(\mathbf{v}, D^-)$;
- 9 $o_{t+1}^1 \leftarrow o_t^1$, $o_{t+1}^2 \leftarrow o_t^2 + 1$, $o_{t+1}^3 \leftarrow \mathbf{v}$;
- 10 The environment is transited to s_{t+1} ;
- 11 Store $(s_t, a_t, r_{t+1}, s_{t+1})$ in to replay buffer $\mathcal{M}$;
- 12 $s_t \leftarrow s_{t+1}$;
- 13 **end**

selection is regarded as a sequential decision-making process, and Algorithm 1 illustrates this methodology. Given a dataset D^- , the agent explores the feature space in a sequential scanning order. When scanning a feature, the agent is required to employ the neural network to make decisions regarding the selection or deselection of said features (lines 5–13). The state s_t consists of three components: the dataset representation o_t^1 generated by the GCN-based state representation method, the sequence number of the feature being decided o_t^2 , and the current feature subset o_t^3 . These components are concatenated to form a unified input for the neural network, while each distinct output of the network represents the Q-values associated with selection or deselection actions.

The ϵ – Greedy algorithm is utilized to enable the agent to balance exploration and exploitation (line 6). The agent randomly selects an action with a significantly lower probability $\epsilon \in [0, 1]$ and the action with the highest Q-values with a higher probability $1 - \epsilon$. After performing a selection or deselection action on the currently selected feature subset, a reward signal r_{t+1} is generated based on the classification accuracy (lines 7–8), and subsequently, the environment transitions to the next state s_{t+1} (lines 9–10). Finally, the interactive data during a time slot $(s_t, a_t, r_{t+1}, s_{t+1})$ will be stored in the replay buffer $\mathcal{M}$ for the training stage (lines 11).

D. DQN Based Iterative Training

1) **DQN Algorithm:** This paper employs the DQN algorithm to perform end-to-end learning of the parameters in the decision network Q . DQN [31] is a model-free, off-policy RL algorithm with sample efficiency and learning stability. DQN employs a parameterized state-action value function $Q(s_t, a_t|\theta)$ to adopt the continuous state domains. The parameters of the Q-function are trained by minimizing the following mean-squared temporal

Algorithm 2: The Procedure of the Iterative Training.

Input: Training dataset: $D \in \mathbb{R}^{m \times n}$; capacity of replay memory: P ; size of the sampled data: b .

Output: A learned decision network Q .

- 1 Initialize replay memory $\mathcal{M}$ with capacity P ;
- 2 Initialize the parameter of value network Q and target network $\hat{Q}$ respectively;
- 3 **for** $\text{episode} = 1$ **to** K **do**
- 4 Sample a balanced data $D^- \in \mathbb{R}^{b \times n}$ from D using under-sampling technique;
- 5 Update $\mathcal{M}$ by neural network based feature selection process (Algorithm 1);
- 6 Randomly sample a mini-batch transitions $(s_t, a_t, r_{t+1}, s_{t+1})$ from replay buffer $\mathcal{M}$;
- 7 Update Q-network $\theta \leftarrow \theta - \alpha \hat{\nabla} \mathcal{L}(\theta)$;
- 8 Update target network $\theta' \leftarrow \tau \theta + (1 - \tau) \theta'$;
- 9 **end**

difference (TD) error:

$$\mathcal{L}(\theta) = \mathbb{E}_{s_t, a_t, s_{t+1}} \left[\frac{1}{2} (Q(s_t, a_t|\theta) - y)^2 \right] \quad (7)$$

with

$$y = r_{t+1} + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1}|\theta'), \quad (8)$$

where $\hat{Q}(s_{t+1}, a_{t+1}|\theta')$ is the target Q-function whose parameters are exponentially updated towards the Q-function for stability purposes. The parameters of the Q-function θ can be updated by the gradient descent method:

$$\begin{aligned} \hat{\nabla} \mathcal{L}(\theta) = & \nabla_{\theta} Q(s_t, a_t|\theta) [r_{t+1} + \gamma \max_{a_{t+1}} \hat{Q}(s_{t+1}, a_{t+1}|\theta') \\ & - Q(s_t, a_t|\theta)]. \end{aligned} \quad (9)$$

2) **Iterative Training Process:** The procedure of the proposed iterative training method is illustrated in Algorithm 2. The training episode starts with sampling a balanced dataset from the original imbalanced dataset, which serves as the environment for that particular episode (line 4). The agent then proceeds to make sequential decisions on the feature selection problem and update the replay buffer $\mathcal{M}$, as illustrated in Algorithm 1.

In addition, the data for the agent learning is generated sequentially through interactions, indicating that these samples do not follow an independent and identically distributed pattern as assumed by most ML methods. Therefore, the experience replay memory technique is adopted. The state, action, reward, and new state during a time slot $(s_t, a_t, r_{t+1}, s_{t+1})$ will be stored in the replay buffer $\mathcal{M}$. During the training process, a mini-batch data is sampled from $\mathcal{M}$ randomly (line 6). The parameters of the network Q are updated via (7)–(9). And the parameters of the target network are updated as follows [32]:

$$\theta' \leftarrow \tau \theta + (1 - \tau) \theta', \quad (10)$$

where $\tau \ll 1$ is the learning rate.

Algorithm 3: Feature weight calculation process.

Input: The learned network: Q ; training dataset: $D \in \mathbb{R}^{m \times n}$; size of the sampled data: b ; binary encoding $v \in \mathbb{R}^n$; weight vector $w \in \mathbb{R}^n$.

Output: The weight vector w .

```

1 Initialize weight vector  $w$  with zero;
2 for  $episode = 1$  to  $N$  do
3   Initialize binary encoding  $v$  with zero;
4   Sample a balanced data  $D^- \in \mathbb{R}^{b \times n}$  from  $D$  using
     under-sampling technique;
5    $s_t \leftarrow \text{concatenate } \langle o_t^1, o_t^2, o_t^3 \rangle$ ;
6    $\text{Baseline} \leftarrow \text{Acc}(D^-)$ ;
7   for  $t = 1$  to  $n$  do
8      $v_t = \text{argmax}_{a_t} Q(s_t, a_t)$ ;
9     The environment is transited to  $s_{t+1}$ ;
10     $s_t \leftarrow s_{t+1}$ ;
11  end
12   $w \leftarrow w + v \cdot \frac{\text{Acc}(v, D^-) - \text{Baseline}}{\text{Baseline}}$ 
13 end

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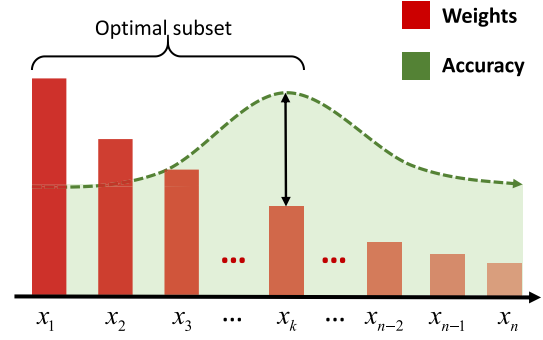


Fig. 3. The forward-selection process. The features are ranked in descending order based on the weights. A feature subset starts as an empty set and gradually adds one feature at a time, evaluating the classification accuracy after each addition. The subset that yields the highest accuracy is deemed the ultimate result.

selection process involves iteratively adding features to a subset and calculating the corresponding classification accuracy on the validation dataset. The optimal feature subset is determined by selecting the highest accuracy.

E. Weighted Model

After the training stage, it remains essential to determine the final optimal feature subset based on the trained decision network for practical classification applications. To this end, a weighted model is proposed in which the decision network is employed to generate weight for each feature, subsequently leading to identifying an optimal feature subset based on these weights.

1) *Calculate Feature Weight:* The trained decision network will conduct feature selection on supplementary datasets and synthesize the outcomes of this process. The process of the weighted model is illustrated in Algorithm 3. The trained Q network is directly employed on additional epochs to identify the corresponding optimal feature subset (lines 3–12). At the beginning of each episode, the accuracy of datasets without feature selection is recorded as a baseline (line 6). Then, the agent conducts feature selection by leveraging the trained decision network Q (lines 7–11). At the end of this episode, the weight vector is updated with the following rule:

$$w \leftarrow w + v \cdot \frac{\text{Acc}(v, D^-) - \text{Baseline}}{\text{Baseline}}, \quad (11)$$

where $\text{Acc}(v, D^-)$ denotes the accuracy on datasets after feature selection. The weight update is only applied to the selected features, and the magnitude of the update is determined by the extent to which the feature subset enhances classification accuracy. The weight vector is robustly enhanced through continuous updates on diverse datasets, aiming to mitigate the influence of randomness from the under-sampling step. This process ensures that only those features which exhibit relevance across a significant proportion of datasets are retained.

2) *Forward Selection Process:* The final feature subset is determined using a simple forward-selection method based on the weights, as illustrated in Fig. 3. The features are initially sorted in descending order based on the weights. Then, the feature

V. EXPERIMENTS

In this section, the proposed RLFS method is extensively evaluated through experiments in the context of breast cancer iDEs prediction problem. The experimental setup is first introduced, including the data information and parameter settings. Several critical parameters influencing the algorithm are examined in detail. Then, the experimental results of the RLFS in breast cancer iDEs prediction problem will be presented, including the convergence of RLFS and the analysis of feature selection results. Finally, the RLFS is compared with several state-of-the-art FS approaches, and a data balancing technique to demonstrate the superiority of the proposed method. The following sections describe the experiments in detail.

A. Experimental Settings

1) *Data Materials:* The Medical Ethics Committee of the Xiangya Hospital of Central South University approved this study (No. 2023–11237). The simulation study assembled a large-scale cohort dataset with 2645 patients from the Department of Breast Surgery, Xiangya Hospital, Changsha, China. The analyzed experimental data in this paper pertain to a cohort of patients who were pathologically diagnosed with primary breast cancer before surgery and received surgical treatment for breast cancer in Xiangya Hospital from 2011 to 2015. The follow-up period of less than five years resulted in the exclusion of several patients. The final analysis of 5-year iDEs included 2104 patients, comprising 1647 iDE cases and 430 non-iDE cases, as illustrated in Fig. 4. The entire dataset is divided into three groups, namely the training set (1052 patients), validation set (526 patients), and testing set (526 patients). The training set is utilized for RL model training purposes; the validation set is employed to determine the optimal feature subset using the weighted model; the testing set is used to evaluate the performance of the proposed FS method.

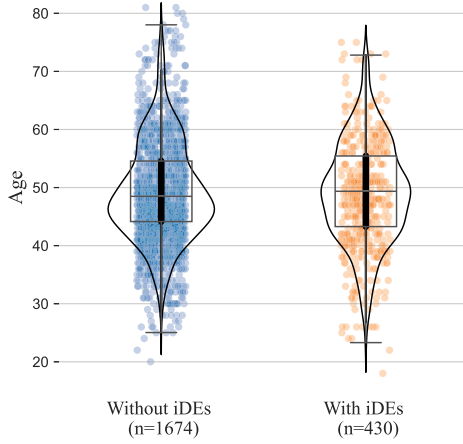


Fig. 4. The age distributions of between patients with iDEs and non-iDEs.

TABLE I
THE PARAMETER CONFIGURATIONS OF THE DQN ALGORITHM

Parameter	Value
replay buffer capacity P	4096
batch size	64
discount factor γ	0.99
learning rate τ	0.01
training interval	10
exploration probability ϵ	0.5-0.05

The cohort has documented 25 clinical features associated with the primary breast tumor. Demographic and genetic features include age and body mass index (BMI). Treatment features include breast-conserving surgery (BCS) and medicine before surgery (MBS). Clinicopathological features include pathological subtype (Path), histological grade (Hist), nodes status, T-stage, tumor size, N-stage, TNM-staging, estrogen receptor (ER) status, progesterone receptor (PR) status, human epidermal growth factor receptor 2 (HER2) status, Ki-67, molecular subtype (Mole). Blood routine examination features include basophil count (BAC), hematocrit value (HCT), lymphocyte percentage (LYP), mean corpuscular hemoglobin (MCH), monocyte percentage (MOP), mean platelet volume (MPV), platelet hematocrit (PTC), platelet count (PLT), red blood cell count (RBC).

2) Parameter Settings: In this experiment, the sampling quantities of the balanced dataset in each iteration is set to 200 ($b = 200$). The auto-encoder is a single-layer neural network that reduces dimensionality from 200 to 128. The GCN is built by a two-layer neural network consisting of 128 and 64 nodes in each layer. The Q network is a two-layer neural network with 64 and 32 nodes. The activate function of all networks is set as ReLU. The Adam optimizer is employed to optimize the parameters of all neural networks while employing a rate of 0.025 with linear attenuation. The parameters of the DQN algorithm are summarized in Table I, where the training interval denotes the amount of data accumulated before updating the network parameters. The k -Nearest Neighborhood (k -NN) algorithm is employed as a classifier to evaluate the candidate feature subset,

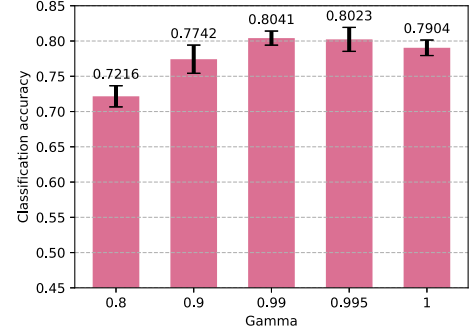


Fig. 5. Performance comparison of different discount factors.

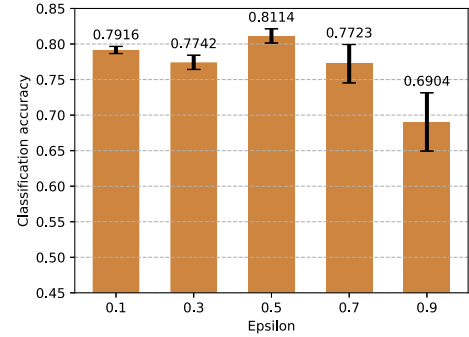


Fig. 6. Performance comparison of different exploration probabilities.

with the parameter k set to 1. In order to mitigate the impact of dataset division, a 5-fold cross-validation approach is utilized to return the reward of the action. The proposed algorithm is executed independently 10 times with different random seeds in order to mitigate the impact of randomness. For each run, the RL models undergo training for 4000 episodes, with an additional 50 episodes dedicated to generate the weighted model.

B. The Impact of Key Parameters

In this section, several parameters which greatly influence the algorithm's performance are discussed separately.

1) Discount Factor: Firstly, the discount rate γ in RL represents the effect on future expected rewards. Fig. 5 illustrates the classification accuracy of the RLFS under various discount rates. Note that a relatively small γ represents that the agent is solely influenced by immediate rewards, leading to its shortsighted behavior and consequently diminishing its effectiveness. On the other hand, due to the random generation of the dataset in each iteration, the training environment remains inherently uncertain. Therefore, an excessively high γ leads to difficulty in algorithm convergence. Thus, the discount rate γ in this paper is set to 0.99.

2) Exploration Probability: The ϵ - Greedy mechanism balances exploration and exploitation during decision-making. In this paper, the probability of exploration decreases linearly from an initial value ϵ_0 to 0.05. Fig. 6 shows the classification accuracy of the RLFS under various initial epsilon values. The observation from Fig. 6 reveals that the algorithm necessitates a relatively high initial epsilon due to the complexity of the environment.

TABLE II

THE NUMBER OF EPOCHS THAT REACH CONVERGENCE AND ACCURACY PERFORMANCE UNDER DIFFERENT SAMPLING QUANTITIES

Criteria	Sampling quantity b			
	50	100	200	400
Epochs	-	10000	4000	1500
Acc	-	0.8125 ± 0.0378	0.8118 ± 0.0164	0.7874 ± 0.0259

However, an excessive exploration probability may induce instability in the training process, leading to significant variance in the output. Thus, the initial epsilon ϵ_0 is set to 0.5.

3) *Sampling Quantity*: The sampling quantity represents the size of the balanced dataset generated in each epoch. Table II illustrates the number of epochs required for convergence and average prediction accuracy under different sample quantities. It can be observed from Table II that the number of epochs required to achieve convergence increases as the sample quantities decrease. The algorithm fails to converge when $b = 50$. The limited volume of data results in the sampled dataset representing only a tiny fraction of the data space, thereby posing challenges for algorithm convergence. A large sampling quantity, on the other hand, can expedite training time as the neural network can efficiently learn from more data at once. However, excessive data utilization in each epoch reduces dataset diversity, consequently diminishing the model's generalization capability. Therefore, the sampling quantity is set to 200.

C. Algorithm Performance

1) *Convergence Analysis*: The accuracy of each epoch cannot accurately reflect the performance of the decision network due to variations in the data used across epochs. Therefore, a metric referred to as *improvement ratio* R is employed to assess the convergence of the algorithm. The improvement ratio of the feature subset $\mathbf{v}$ determined by the decision network to the sampled dataset D^- in a specific epoch is defined as

$$R(\mathbf{v}, D^-) = \frac{Acc(\mathbf{v}, D^-)}{Acc(D^-)}, \quad (12)$$

where $Acc(\mathbf{v}, D^-)$ and $Acc(D^-)$ represent the classification accuracy with and without FS, respectively. If the value of R is greater than 1, it indicates that the classification performance improves after the neural network-based feature selection; otherwise, it suggests a decline in performance.

The convergence curve of the improvement ratio is illustrated in Fig. 7, where the red dashed line is the baseline that represents the $R = 1$; the shadow areas denote the standard deviations of the improvement ratio. It can be observed from Fig. 7 that the average improvement ratio of the proposed method keeps increasing as the training process continues. The improvement ratio R of nearly all episodes surpasses 1 and consistently ranges between 1.05–1.10 in the later iteration of the algorithm. The implication is that the RL algorithm's agent can acquire a robust FS policy, which can significantly enhance classification performance on different datasets.

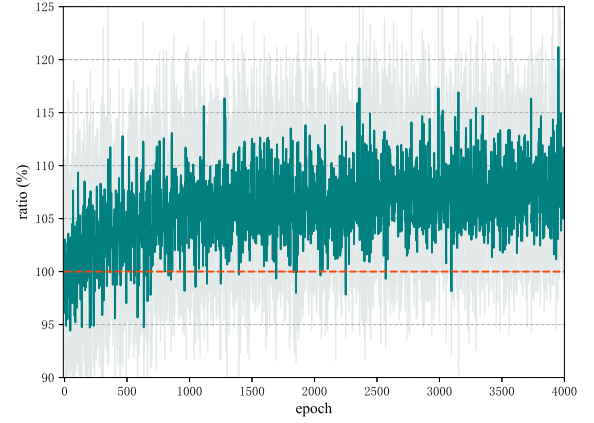


Fig. 7. The convergence curve of the reinforcement learning. The red line represents a baseline, representing the results of not performing feature selection.

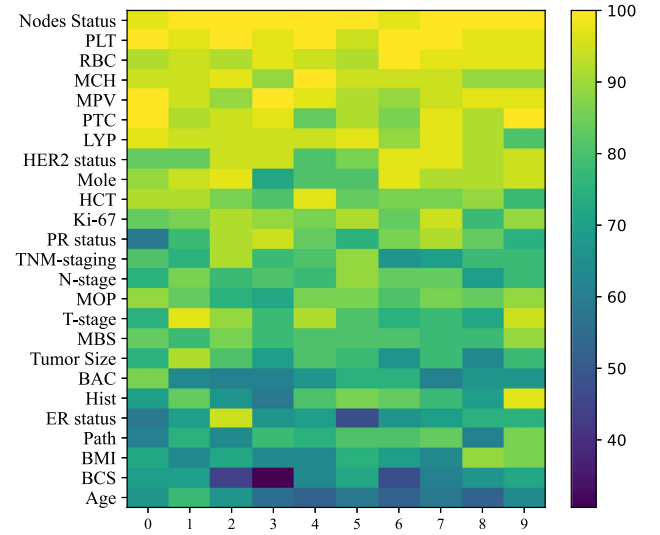


Fig. 8. The statistical results of the features in the weighted model. Each column represents a separate experiment, and each row represents the frequency at which each feature was selected in a single experiment.

2) *Feature Selection Analysis*: The statistical frequency of the features selected by the trained agent across each of the ten independent training with 50 episodes is illustrated in Fig. 8. The weight distribution of each feature is visually represented by the boxplot diagram as illustrated in Fig. 9. It can be observed that the most crucial feature for 5-year iDEs prediction is Nodes States (with a nearly 100% statistical frequency percentage for all episodes), which represents the number of metastatic lymph nodes. This feature has been scientifically proven to be a prognostic indicator for breast cancer recurrence [33]. Other significant clinicopathological features include HER2 status, molecular subtype, and Ki-67 status (with a 70% statistical frequency percentage for all episodes), which are well recognized for their crucial role in guiding therapeutic decision-making [34]. In addition, the blood routine examination features are found to be significantly associated with the prognosis of breast cancer (with a 90% statistical frequency percentage for all episodes), including platelet count (PLT), red blood cell count (RBC),

TABLE III

COMPARISON OF PROPOSED METHOD AND OTHER ALGORITHMS IN TERMS OF CLASSIFICATION PERFORMANCE AND THE NUMBER OF THE SELECTED FEATURES d

Criteria	SFE-PSO Mean $\pm$ Std	HFSIA Mean $\pm$ Std	SM-MOEA Mean $\pm$ Std	SaWDE Mean $\pm$ Std	MARLFS Mean $\pm$ Std	RLFS Mean $\pm$ Std
d	11.7 $\pm$ 3.2	9.6 $\pm$ 4.7	7.4 $\pm$ 2.4	12.0 $\pm$ 2.5	11.2 $\pm$ 3.8	8.0 $\pm$ 1.2
Accuracy	0.7946 $\pm$ 0.0271	0.7862 $\pm$ 0.0375	0.7694 $\pm$ 0.0186	0.7914 $\pm$ 0.0271	0.7861 $\pm$ 0.0298	0.8118 $\pm$ 0.0164
Precision	0.6323 $\pm$ 0.0246	0.6243 $\pm$ 0.0321	0.6125 $\pm$ 0.0258	0.6314 $\pm$ 0.0264	0.6226 $\pm$ 0.0285	0.6400 $\pm$ 0.0195
Recall	0.3678 $\pm$ 0.0274	0.3568 $\pm$ 0.0317	0.3468 $\pm$ 0.0268	0.3541 $\pm$ 0.0240	0.3601 $\pm$ 0.0276	0.4243 $\pm$ 0.0196
F1-score	0.5000 $\pm$ 0.0260	0.4906 $\pm$ 0.0319	0.4796 $\pm$ 0.0263	0.4927 $\pm$ 0.0252	0.4913 $\pm$ 0.0280	0.5104 $\pm$ 0.0195

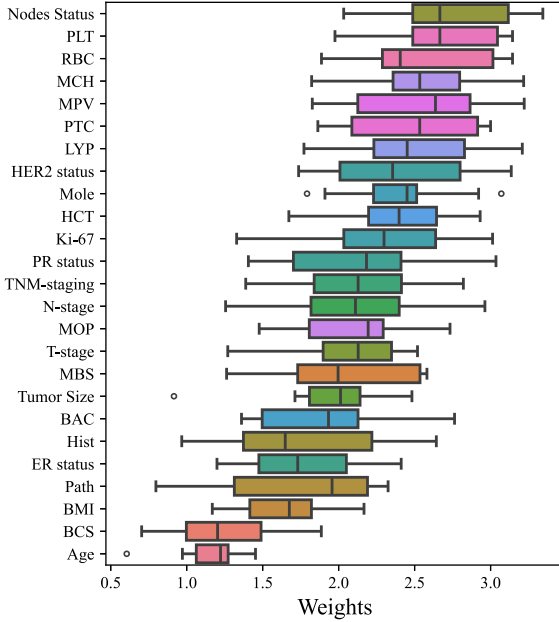


Fig. 9. The distribution of each feature weight calculated in weighted model.

mean corpuscular hemoglobin (MCH), mean platelet volume (MPV), platelet hematocrit (PTC), and lymphocyte percentage (LYP), which is an intriguing discovery. This suggests that blood routine, as a common clinical test parameter, has a potential association with breast cancer prognosis, aligning with the results of several studies [35].

D. Comparing to State-of-The-Art Methods

To evaluate the performance of the proposed method, the RLFS is compared with that of several recently state-of-the-art FS methods, including SFE-PSO [36], HFSIA [37], SM-MOEA [38], SaWDE [39], and MARLFS [40]. The maximum number of iterations for each algorithm is 4000. The parameters of these algorithms are set according to the settings in their original papers. The following standard metrics are employed to assess the efficacy of FS algorithms, including *Accuracy*, *Precision*, *Recall*, and *F1-score*:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$Precision = \frac{TP}{TP + FP} \quad (14)$$

TABLE IV

COMPARISON OF PROPOSED METHOD AND OTHER ALGORITHMS IN TERMS OF RUNNING TIME

	SFE-PSO	HFSIA	SM-MOEA	SaWDE	MARLFS	RLFS
Time	1461.5s	1274.7s	1924.3s	1740.7s	2137.1s	1232.4s

$$Recall = \frac{TP}{TP + FN} \quad (15)$$

where TP , TN , FP , FN denote true positive, true negative, false positive and false negative. $F1\text{-score} = 2 \times P \times R / (P + R)$, where P and R represent *precision* and *recall*, respectively.

1) *The Performance of Feature Selection*: Table III presents the comparison of feature selection results, where d represents the average size of selected features. The proposed RLFS algorithm achieves higher accuracy compared to all state-of-the-art FS algorithms. This accomplishment is attained through the well-crafted FS framework, which leverages the RL algorithm. Specifically, employing classification accuracy as a reward signal can effectively direct the agent toward generating more optimal feature subsets. In addition, the RLFS significantly outperforms other algorithms regarding other classification metrics, especially recall rate. The reason behind this is that the RL agent is trained on balanced datasets, which are constructed using the resampling technique. As a result, the selected features more accurately represent the information of the minority class. The low standard deviation also serves as evidence of the stability and robustness of RLFS. This is mainly due to the iterative training mechanism and the weighted model proposed in this paper. The data utilized in each iteration varies, enabling the agent to evolve into a robust decision network throughout the iterative process. The weighted model integrates various FS outcomes to yield a final feature subset, producing a more robust and stable result.

2) *The Performance of Computational Time*: Table IV shows the average running time of each FS algorithm. The running time of RLFS is observed to be lower than that of other FS algorithms. Evaluating the feature subset by ML classifiers constitutes the most time-consuming aspect of FS algorithms. In contrast to competitors, the RLFS employs a reduced dataset generated through under-sampling technique instead of the entire dataset for evaluating the feature subset in each iteration. This strategy ensures a relatively efficient runtime for RLFS. In addition, the MARLFS employs a multi-agents RL framework to search the optimal feature subset, where each agent represents an individual feature. The simultaneous training of numerous neural networks needs substantial computational resources. In contrast, this paper

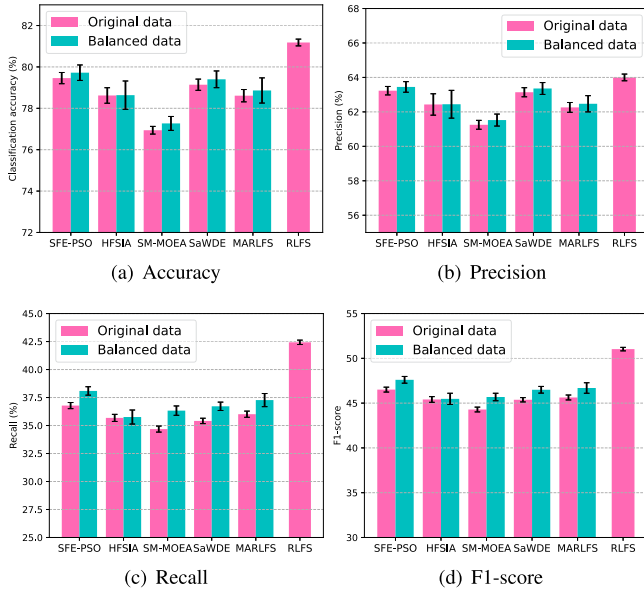


Fig. 10. Comparison with the state-of-the-art FS methods that incorporate data balancing technique.

formulates the FS problem as a Markov decision process, which requires only a single network, thereby enhancing efficiency.

3) *The Performance With Data Balancing Technique:* The above algorithms only address the FS problem while ignoring data imbalance problem. To further evaluate the performance of the proposed method in terms of FS for imbalanced data, the RLFS is compared with these algorithms that incorporate data balancing technique. Specifically, all the competitors employ the FS after data resampling (DS+FS) framework [20], where the well-received resampling technique, SMOTE [18], is utilized to generate balanced dataset before applying FS algorithms. Fig. 10 shows the results of the comparative experiment. Although the competitors' performance combined with the resampling technique has been improved, it remains inferior to that of the proposed algorithm. This occurs due to the limitation of a singular constructed dataset in adequately capturing the true data space, which may result in the algorithm converging at different local optima. In addition, it can be observed that the resampling technique introduces randomness, leading to an increase in the standard deviation. In conclusion, the algorithm presented in this study demonstrates superior global stability and robustness compared to existing algorithms.

E. Comparison With Different Classifiers

The predictive performance depends not only on FS methods but also on utilizing ML classifiers. This section evaluates the proposed FS method for its consistency and superiority over other baseline methods on multiple ML classifiers, including LASSO, Decision Tree (DT), Support Vector Machines (SVM), and XGBoost. The overall accuracy of various FS methods with different ML classifiers is presented in Table V. The results illustrate that the proposed method attains the highest prediction accuracy among all ML classifiers. Specifically, the

TABLE V
OVERALL ACCURACY OF DIFFERENT FS METHODS WITH DIFFERENT ML CLASSIFIERS

Algorithms	Accuracy (%)			
	LASSO	DT	SVM	XGBoost
SFE-PSO	80.41	69.75	79.74	78.52
HFSIA	79.24	68.44	78.41	77.24
SM-MOEA	77.61	66.21	77.36	76.78
SaWDE	80.03	70.26	80.21	79.21
MARLFS	78.74	68.81	79.96	78.23
RLFS	81.28	71.58	81.44	80.37

imbalanced data also poses a challenge to the information gain-based model, DT. It can be observed that the proposed method effectively addresses this issue for DT. Overall, the proposed method demonstrates a higher level of robustness than other ML classifiers.

VI. CONCLUSION

This paper proposes a novel reinforcement learning-based feature selection framework to address the issue of imbalanced datasets in breast cancer iDEs prediction. A decision network is designed to automatically identify the useful features from the given dataset using reinforcement learning algorithm. The decision network's feature selection capacity for imbalanced data is enhanced through an iterative mechanism, wherein a balanced dataset is constructed during each iteration. The proposed method is utilized to predict the iDEs of breast cancer in a clinical patient cohort obtained from Xiangya Hospital of Central South University, China. The analysis of the feature selection results reveals the significance of different features in predicting breast cancer prognosis. Compared to several state-of-the-art feature selection algorithms, the proposed method has enhanced prediction accuracy by over 3% and exhibited superior stability.

The proposed algorithm exhibits potential for application in medical prognostic models characterized by significant class imbalance. However, further investigation is still needed to rectify its existing limitations. For example, under-sampling is a basic sampling technique. Further investigation will be conducted to explore the implementation of alternative advanced techniques. The weighted model employs a greedy approach to the final feature subset, which may become trapped in local optima. Dynamic clustering can be harnessed to optimize the method's performance.

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